Summary 1

Systems of Linear Equations

Contents

1.1 Forming the Equations from the Context and Entering Them into the GC 2
1.2 N Variables and N Independent Equations 3
1.3 Cubic from Two Points, a Turning Point and One More Condition 4
1.4 Two Equations, Three Variables 5

1.1 Forming the Equations from the Context and Entering Them into the GC

Example 1.1.1

Three different blends of coffee beans -- Blend A, Blend B and Blend C -- are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	**Discounts given for each package**			**Total price after the discount**
	Blend A	**Blend B**	**Blend C**	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. **[3]**

(MI 2025 Prelim P1 Q4)

Solution.

(i) Let the usual prices be \$$a$, \$$b$, \$$c$.

$$a + b + c = 176 \quad ---(1)$$
$$0.85a + 0.90b + 0.95c = 161.80 \quad ---(2)$$
$$0.93a + 0.95b + 0.98c = 169.56 \quad ---(3)$$

Using GC, $a = 32,\ b = 44,\ c = 100$

$\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100

1.2 N Variables and N Independent Equations

Two linear equations are independent if and only if neither equation is a linear combination of the other.

Example 1.2.1

A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. **[3]**
Hence, find the values of x_{10} and x_{11}. **[2]**

(HCI 2025 Prelim P1 Q2)

Solution.

$$\begin{aligned} n &= 1 \\ x_2 &= a(-3)^1 + b(1)^2 + cx_1 \\ -8 &= -3a + b + c \qquad \cdots (1) \\ n &= 2 \\ x_3 &= a(-3)^2 + b(2)^2 + cx_2 \\ 70 &= 9a + 4b - 8c \qquad \cdots (2) \\ n &= 3 \\ x_4 &= a(-3)^3 + b(3)^2 + cx_3 \\ -377 &= -27a + 9b + 70c \qquad \cdots (3) \end{aligned}$$

Using GC, $a = 2,\ b = 3,\ c = -5$

$$x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$$

Using GC,

$$x_{10} = -7\,210\,868$$
$$x_{11} = 36\,172\,738$$

1.3 Cubic from Two Points, a Turning Point and One More Condition

Example 1.3.1

A function f is defined by $\mathrm{f}(x) = ax^3 + bx^2 + cx + d$ where a, b, c and d are constants. The graph of $y = \mathrm{f}(x)$ has a turning point at $x = -1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x = 16$, find the values of a, b, c and d. **[4]**

(SAJC 2025 Prelim P1 Q1)

Solution.

$$\begin{aligned}
\mathrm{f}'(x) &= 3ax^2 + 2bx + c \\
\mathrm{f}'(-1) &= 0 \\
3a - 2b + c &= 0 \qquad \text{---(1)} \\
\mathrm{f}(2) &= 10 \\
8a + 4b + 2c + d &= 10 \qquad \text{---(2)} \\
\mathrm{f}(-2) &= -2 \\
-8a + 4b - 2c + d &= -2 \qquad \text{---(3)} \\
\int_{-2}^{2} \left(ax^3 + bx^2 + cx + d\right)\mathrm{d}x &= 16 \\
\left[\tfrac{a}{4}x^4 + \tfrac{b}{3}x^3 + \tfrac{c}{2}x^2 + dx\right]_{-2}^{2} &= 16 \\
(4a + \tfrac{8}{3}b + 2c + 2d) - (4a - \tfrac{8}{3}b + 2c - 2d) &= 16 \\
\tfrac{16}{3}b + 4d &= 16 \qquad \text{---(4)}
\end{aligned}$$

Using GC, $a = 3,\ b = 0,\ c = -9,\ d = 4$

1.4 Two Equations, Three Variables

Example 1.4.1

[Question: two equations, three variables -- not yet sourced from the papers on the Mac]

Solution.

[Solution: to follow]

Summary 2

Inequalities

Contents

2.1 Removing a Square Root or an Always-Positive Factor 7
2.2 Substitution Involving $1/x$ 8
2.3 Solving Graphically 9

2.1 Removing a Square Root or an Always-Positive Factor

Example 2.1.1

Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

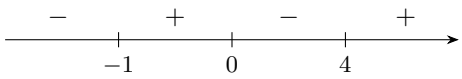
[4]
(ACJC 2026 Prelim P1 Q1)

Solution.

$x \geqslant 3x-6$
$x \leqslant 3 \qquad ---(1)$

$x^2+2x+5 = (x+1)^2+4 > 0$ for all real x

$$\frac{x(5x+9)}{x^2+2x+5} > x$$
$$5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$$
$$x^3-3x^2-4x<0$$
$$x(x-4)(x+1)<0$$



$x<-1$ or $0<x<4 \quad ---(2)$

$(1) \cap (2)$:

$\therefore\ x<-1$ or $0<x\leqslant 3$

2.2 Substitution Involving $1/x$

Example 2.2.1

(i) Solve the inequality $3\left|x^2-4\right| \leqslant |x+2|$. **[4]**

(ii) Hence, solve $\dfrac{3}{x^2}\left|1-4x^2\right| \leqslant \left|\dfrac{1+2x}{x}\right|$. **[2]**

(SAJC 2025 Prelim P2 Q1)

Solution.

(i) $3|x-2||x+2| \leqslant |x+2|$

$x=-2: \quad 0 \leqslant 0$ (true)

$x \neq -2: \ |x+2| > 0$

$3|x-2| \leqslant 1$

$-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$

$\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$

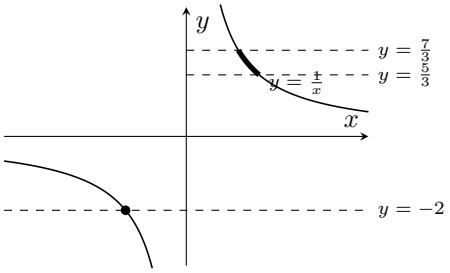
$\therefore \ x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$

(ii) $\dfrac{3}{x^2}\left|1-4x^2\right| = 3\left|\dfrac{1}{x^2}-4\right| \quad (\because x^2>0)$

$\left|\dfrac{1+2x}{x}\right| = \left|\dfrac{1}{x}+2\right|$

$3\left|\left(\dfrac{1}{x}\right)^2-4\right| \leqslant \left|\dfrac{1}{x}+2\right| \quad (x \to \frac{1}{x})$

$\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$



$\therefore \ x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$

2.3 Solving Graphically

Example 2.3.1

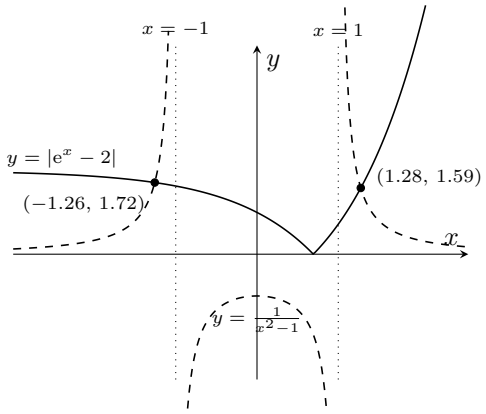
(a) Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. **[3]**

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. **[3]**

(RVHS 2026 Prelim P1 Q2)

Solution.

(a)



Using GC, intersections at $x = -1.2581$, $x = 1.2770$

$\therefore\ x < -1.26 \ \text{ or } \ -1 < x < 1 \ \text{ or } \ x > 1.28$

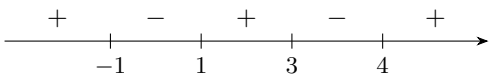
(b)

$$\frac{x+2}{x-1} - \frac{2}{x-3} \geqslant 0$$

$$\frac{(x+2)(x-3) - 2(x-1)}{(x-1)(x-3)} \geqslant 0$$

$$\frac{x^2 - 3x - 4}{(x-1)(x-3)} \geqslant 0$$

$$\frac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$$



$\therefore\ x \leqslant -1 \ \text{ or } \ 1 < x < 3 \ \text{ or } \ x \geqslant 4$